Lab Report

Scenario: Online Food Delivery System

1. Problem Statement

Customers browse restaurants and place orders. Restaurants receive orders and update order status. Delivery riders receive delivery requests and update delivery status. A payment service processes online payments. An administrator manages users, restaurants, and complaints.

2. Extracted Requirements

- R1: Customers must be able to browse restaurants/menus and place orders.
- R2: Customers must be able to pay for orders online.
- R3: Restaurants must be able to receive orders and update order status.
- R4: Delivery riders must be able to receive delivery requests and update delivery status.
- R5: The system must communicate with a payment service to process payments.
- R6: An administrator must be able to manage users, restaurants, and complaints.

3. Identify External Entities

The following entities are outside the boundary of the Online Food Delivery System but interact with it:

External Entity	**Why External?**	**Major Interaction**
Customer	A person using the system, not an internal software component.	Requests restaurant/menu information, places orders, makes payments, receives order status.
Restaurant	An independent business external to the software system.	Receives new orders; sends order status updates.
Delivery Rider	A person outside the software system who fulfils deliveries.	Receives delivery requests; sends delivery status updates.
Payment Service	A separate external payment-processing service.	Receives payment requests; sends payment confirmation.
Administrator	A person managing the platform, not an internal component.	Manages users, restaurants, and complaints.

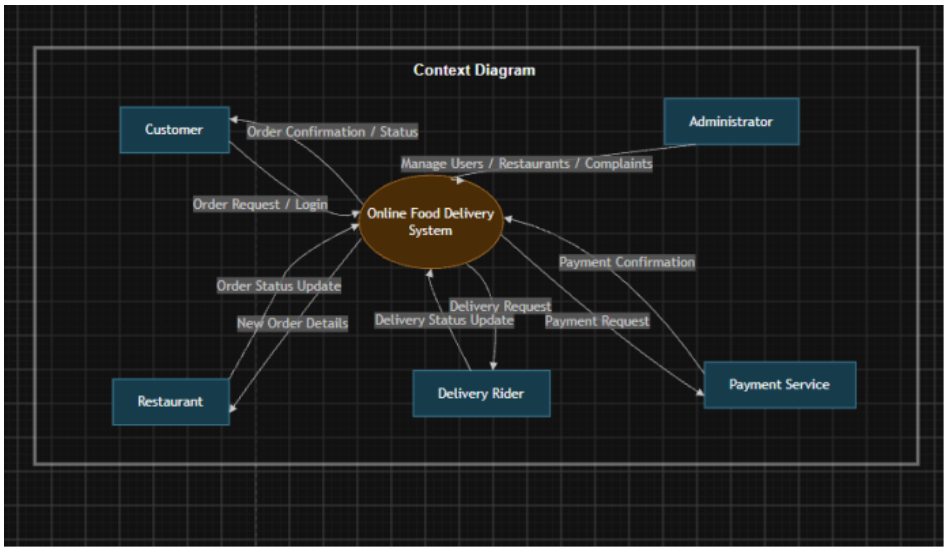
4. Identify Major Information Flow

External Entity	**Information Sent to System**	**Information Received**	**Related Requirements**
Customer	Browse request, order placement request, payment info	Restaurant/menu list, order confirmation, order status updates	R1, R2
Restaurant	Order status update	New order details	R3

Delivery Rider	Delivery status update	Delivery request	R4
Payment Service	Payment confirmation	Payment request	R5
Administrator	Manage users / restaurants / complaints	Management confirmation / records	R6

5. Context Diagram

The context diagram below treats the entire Online Food Delivery System as one unit. Customer, Restaurant, Delivery Rider, Payment Service, and Administrator remain outside the system boundary. Internal modules are intentionally not shown here.



Explanation

- The central node represents the complete software system.
- Customer, Restaurant, Delivery Rider, Payment Service, and Administrator are external entities.
- Flows are labeled with meaningful information (e.g., "Order Request / Login") rather than generic terms such as "data".
- The diagram does not show databases, screens, classes, APIs, or internal modules.
- Payment Service is external because payment processing is performed by a separate service, not by the ordering platform itself.

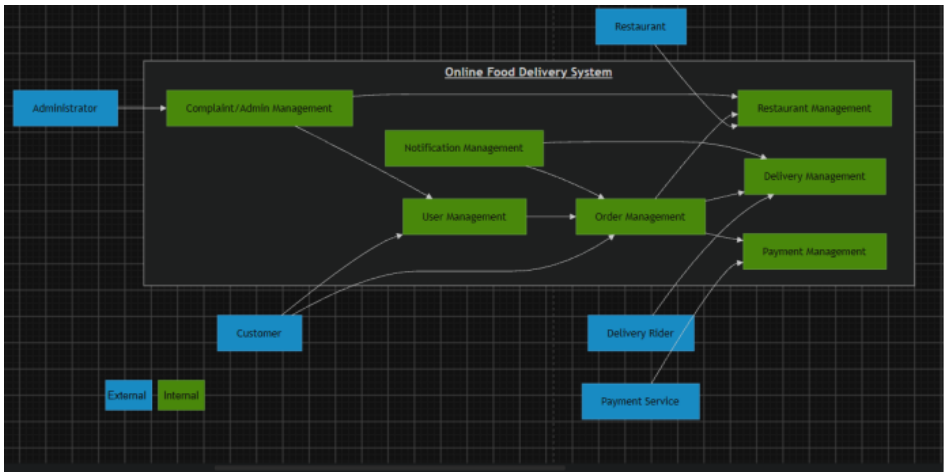
6. Major Internal Modules

Internal Module	**Responsibility**
User Management	Manages customer, restaurant, and rider accounts and profiles.
Restaurant Management	Maintains restaurant listings, menus, and availability.
Order Management	Processes order placement and coordinates order status across modules.

Delivery Management	Assigns delivery requests to riders and tracks delivery status.
Payment Management	Handles payment requests and communicates with the external Payment Service.
Notification Management	Sends status updates and alerts to customers, restaurants, and riders.
Complaint/Admin Management	Allows the administrator to manage users, restaurants, and complaints.

7. System Diagram

The system diagram opens the system boundary and shows the major internal modules and their relationships. It remains high-level and does not describe source code classes or individual database fields.



Explanation

- Order Management is the central coordinating module, connecting to Restaurant Management, Delivery Management, and Payment Management.
- Notification Management supports Order Management and Delivery Management by sending status updates.
- User Management handles account-related requests from Customers and feeds into Order Management.
- Complaint/Admin Management gives the Administrator control over User Management and Restaurant Management (e.g approving/removing restaurants, resolving complaints).
- Restaurant Management, Delivery Management, and Payment Management each interface directly with their corresponding external entity (Restaurant, Delivery Rider, Payment Service).

8. Context Diagram vs. System Diagram

Aspect	**Context Diagram**	**System Diagram**
Perspective	External / boundary view	Internal high-level view
System representation	One complete system	Major internal modules
External entities	Shown	Shown where useful for interaction

		context
Internal modules	Not shown	Shown
Main purpose	Define boundary and external interactions	Explain major internal organization
Level of detail	Very high level	More detailed, but still conceptual

9. Requirement to Diagram Traceability

Requirement	**Context Diagram Evidence**	**System Diagram Evidence**
R1 Browse & order	Customer <-> System (Order Request / Login)	User Management -> Order Management
R2 Online payment	System <-> Payment Service	Order Management -> Payment Management
R3 Restaurant order/status	System <-> Restaurant	Order Management <-> Restaurant Management
R4 Delivery request/status	System <-> Delivery Rider	Order Management / Notification Management <-> Delivery Management
R5 Payment processing	System <-> Payment Service	Payment Management <-> Payment Service
R6 Admin manages users/restaurants/complaints	Administrator <-> System	Complaint/Admin Management <-> User Management, Restaurant Management

10. Ambiguities and Assumptions

Ambiguities

- The problem statement does not specify whether Notification Management is a requirement or an implementation choice, it was added as a supporting module for order status updates.
- The scope of "complaints" is not defined , it is unclear whether complaints can be raised by customers, restaurants, or riders, or all three.

Assumptions

- Payment Service is assumed to be a separate external system.
- Restaurants are assumed to have some interface (app/portal) to receive orders and update status, since the requirement states they "receive orders and update order status".
- Administrator is assumed to be a single internal role with access to Complaint/Admin Management, rather than an external organization.